Rank-Two and Finite-Set Results for the Cylinder Constant in Dymond–Maleva

Automatic Math Discovery Smoke Test

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Source and Target

The source is Michael Dymond and Olga Maleva, *Extreme non-differentiability of typical Lipschitz mappings*, arXiv:2504.04117. In Definition 4.2 they associate to a finite-rank operator $T \in \mathcal{L}(X, Y)$ the constant

$$\mathfrak{C}(T) = \min_{(w_i)_{i=1}^{l}} \max_{1 \le j \le l} \left\| \sum_{i=1}^{j} w_i^* \circ T(\cdot)\, w_i \right\|_{\mathrm{op}},$$

where $w_1, \dots, w_l$ is a basis of $T(X)$ and $(w_i^*)_{i=1}^{l}$ is the dual basis. Lemma 4.4 then constructs, for a compact purely unrectifiable set E in a finite-dimensional domain, a small supported map $g : X \to T(X)$ with derivative approximately T near E and

$$\mathrm{Lip}(g) \le \mathfrak{C}(T) + \theta.$$

Immediately before Definition 4.2, the authors ask whether this upper bound is optimal.

this is what Lemma 4.4 achieves; see Remark 4.3. However, in the general setting, we identify a constant $\mathfrak{C}(T) \geq \|T\|_{\mathrm{op}}$, so that the Lipschitz constant of g may be bounded above by $\mathfrak{C}(T)+\theta$. It would be of interest to determine whether this upper bound is optimal.

Definition 4.2. *Let X and Y be normed vector spaces and $T \in \mathcal{L}(X,Y)$ be a nonzero bounded linear operator of finite rank l. We associate to T the constant*

$$\mathfrak{C}(T) := \min\left\{ \max_{1\leq j\leq l} \left\| \sum_{i=1}^{j} \mathbf{w}_i^* \circ T(\cdot)\,\mathbf{w}_i \right\|_{\mathrm{op}} : \mathbf{w}_1,\dots,\mathbf{w}_l \text{ is a basis of } T(X) \right\}.$$

In the above $\mathbf{w}_1^*,\dots,\mathbf{w}_l^* \in T(X)^*$ is the basis of $T(X)^*$ dual to the basis $\mathbf{w}_1,\dots,\mathbf{w}_l$ of $T(X)$ so that, in particular, $T = \sum_{i=1}^{l} \mathbf{w}_i^* \circ T(\cdot)\,\mathbf{w}_i$. We also define $\mathfrak{C}(\mathbf{0}_{\mathcal{L}(X,Y)}) = 0$.

Remark 4.3. *(i) The constant $\mathfrak{C}(T)$ is well-defined, that is, the minimum of the above set exists, due to Lemma C.1.*

(ii) The identity $T = \sum_{i=1}^{l} (\mathbf{w}_i^* \circ T)(\cdot)\,\mathbf{w}_i$ for any choice of $\mathbf{w}_1,\dots,\mathbf{w}_l$ and $\mathbf{w}_1^*,\dots,\mathbf{w}_l^*$ as above implies $\mathfrak{C}(T) \geq \|T\|_{\mathrm{op}}$.

(iii) For any normed space X and $(Y, \|\cdot\|_Y) = (\ell_p, \|\cdot\|_q)$, $1 \leq p \leq q \leq \infty$ (in particular, Hilbert and finite-dimensional Euclidean), and for every $T \in \mathcal{L}(X,Y)$, one has $\mathfrak{C}(T) = \|T\|$. However for any Y of dimension at least 3 there is a norm $\|\cdot\|_Y$ so that for $X = Y$ and $\|\cdot\|_X = \|\cdot\|_Y$ one has $\mathfrak{C}(\mathrm{Id}_X) > 1$.

Lemma 4.4. *Let X and Y be normed spaces, where X is finite-dimensional. Let $E \subseteq U \subseteq X$ be sets, where E is compact and purely unrectifiable and U is open, $\theta > 0$, $T \in \mathcal{L}(X,Y)$. Then there exist a Lipschitz mapping $g\colon X \to T(X)$ and an open subset H of X such that the following statements hold:*

(a) $\operatorname{supp} g \subseteq U$ *and* $Dg(\mathbf{x}) = 0$ *for all* $\mathbf{x} \in X \setminus U$.
(b) $\sup_{\mathbf{x}\in X} \|g(\mathbf{x})\|_Y \leq \theta$.
(c) $E \subseteq H \subseteq \overline{H} \subseteq U$.
(d) $\|Dg(\mathbf{x}) - T\|_{\mathrm{op}} \leq \theta$ *for Lebesgue almost all* $\mathbf{x} \in H$.
(e) $\mathrm{Lip}(g) \leq \mathfrak{C}(T) + \theta$, *where the constant* $\mathfrak{C}(T)$ *is given by Definition 4.2.*

Results

We settle the rank-one and rank-two cases of the optimality question. In this range the cylinder constant always collapses to the trivial lower bound $\|T\|_{\mathrm{op}}$. Consequently the source lemma already gives the hoped-for constant $\|T\|_{\mathrm{op}} + \theta$. If $E \neq \emptyset$, no smaller asymptotic constant can satisfy the derivative requirement.

We also show that $\mathfrak{C}(T)$ is not the pointwise optimal constant for every individual instance of Lemma 4.4. If the compact purely unrectifiable set is finite, then for every finite-rank T, in every rank, one can achieve $\mathrm{Lip}(g) \leq \|T\|_{\mathrm{op}} + \varepsilon$. Since the source paper notes that $\mathfrak{C}(\mathrm{Id}_X) > 1 = \|\mathrm{Id}_X\|$ for some norms in dimension at least 3, finite sets, already singletons, give counterexamples to the literal reading that $\mathfrak{C}(T)$ is always the pointwise optimal constant.

Proof Intuition

The constant $\mathfrak{C}(T)$ is a finite-dimensional basis constant for the range of T, measured after composing the partial-sum projections with T. In rank two there is only one nontrivial partial sum before the full operator. Choose the first basis vector w_1 to be a unit vector in $T(X)$, and choose a norm-one functional w_1^* that norms it. The kernel of w_1^* is a complementary line; taking w_2 in that line makes w_1^* the first coordinate functional. The first partial-sum operator is then

$$x \mapsto w_1^*(Tx)w_1,$$

whose norm is at most $\|T\|_{\mathrm{op}}$, while the second partial sum is exactly T. Since $\mathfrak{C}(T) \geq \|T\|_{\mathrm{op}}$ for every basis, equality follows.

The finite-set construction uses a different idea. Around each point a_m, there is no geometric interaction provided the supporting balls are chosen much smaller than their mutual separation. Choose disjoint tiny balls $B(a_m, R) \subset U$, and an even tinier radius $r \ll R$. On each ball the logarithmic radial cut-off

$$F_m(x) = \beta(\|x - a_m\|)(x - a_m)$$

equals $x - a_m$ on $B(a_m, r)$, vanishes outside $B(a_m, R)$, and can be made $(1+\varepsilon)$-Lipschitz by taking r/R sufficiently small. The balls are so small that jumps between different balls are harmless. Then $g = T \circ F$, where F agrees with F_m on the m-th ball and is zero elsewhere, has derivative T near every point of E, vanishes outside U, has arbitrarily small sup norm, and has Lipschitz constant arbitrarily close to $\|T\|_{\mathrm{op}}$.

Theorem 1 (Rank-two collapse of the cylinder constant)**.** *Let X and Y be normed spaces and let $T \in \mathcal{L}(X, Y)$ have finite rank at most 2. Then*

$$\mathfrak{C}(T) = \|T\|_{\mathrm{op}}.$$

Proof. The case $T = 0$ is part of the definition of $\mathfrak{C}$. If $\operatorname{rank} T = 1$, there is only one basis vector in $T(X)$, and the only partial sum in the definition of $\mathfrak{C}(T)$ is T itself. Thus $\mathfrak{C}(T) = \|T\|_{\mathrm{op}}$.

Assume now that $\operatorname{rank} T = 2$, and set $Z = T(X) \subseteq Y$. Choose $w_1 \in Z$ with $\|w_1\|_Y = 1$. By Hahn–Banach, choose $\varphi \in Z^*$ with $\|\varphi\|_{Z^*} = 1$ and $\varphi(w_1) = 1$. Since $\dim Z = 2$, the kernel of φ is a one-dimensional subspace. Pick any nonzero $w_2 \in \ker \varphi$. Then w_1, w_2 is a basis of Z, and the first functional in the associated dual basis is $w_1^* = \varphi$.

For this basis, the first partial-sum operator is

$$P_1 T : X \to Y, \qquad P_1 T(x) = \varphi(Tx)\, w_1.$$

Therefore

$$\|P_1 T\|_{\mathrm{op}} \leq \|\varphi\|_{Z^*}\ \|T\|_{\mathrm{op}}\ \|w_1\|_Y = \|T\|_{\mathrm{op}}.$$

The second partial sum is T. Hence this basis gives

$$\max\{\|P_1 T\|_{\mathrm{op}}, \|T\|_{\mathrm{op}}\} \leq \|T\|_{\mathrm{op}}.$$

The reverse inequality $\mathfrak{C}(T) \geq \|T\|_{\mathrm{op}}$ is the general observation recorded in Remark 4.3(ii) of the source paper, since the final partial sum is always T. Thus $\mathfrak{C}(T) = \|T\|_{\mathrm{op}}$. □

Theorem 2 (Finite-set prescribed-derivative constant)**.** *Let X be a finite-dimensional normed space, let Y be a normed space, let $T \in \mathcal{L}(X, Y)$ have finite rank, and let $E \subseteq U \subseteq X$, where E is finite and U is open. For every $\eta > 0$ and every $M > 0$, there are an open set H with $E \subseteq H$, and a Lipschitz mapping $g : X \to T(X)$, such that*

$$\operatorname{supp} g \subseteq U, \qquad \sup_{x \in X} \|g(x)\|_Y \leq M, \qquad Dg(x) = T \quad (x \in H),$$

$$Dg(x) = 0 \quad (x \in X \setminus U), \qquad \operatorname{Lip}(g) \leq \|T\|_{\mathrm{op}} + \eta.$$

Proof. If $T = 0$ or $E = \emptyset$, take $g = 0$. Assume $T \neq 0$, and write $E = \{a_1, \dots, a_N\}$. If $N = 1$, put $\Delta = 1$. If $N \geq 2$, put

$$\Delta = \min_{m \neq n} \|a_m - a_n\|_X > 0.$$

Choose $R > 0$ so small that the closed balls $\overline{B}(a_m, R)$ are contained in U, and

$$R < \Delta/4, \qquad \|T\|_{\mathrm{op}} R \leq M.$$

Choose $L > 0$ with

$$\frac{\|T\|_{\mathrm{op}}}{L} \leq \eta,$$

and set $r = Re^{-L}$. Define $\beta : [0, \infty) \to [0, 1]$ by

$$\beta(t) = \begin{cases} 1, & 0 \leq t \leq r, \\ \dfrac{\log(R/t)}{\log(R/r)}, & r < t < R, \\ 0, & t \geq R. \end{cases}$$

For each m, define

$$F_m(x) = \beta(\|x - a_m\|)(x - a_m).$$

Since the balls $B(a_m, R)$ are pairwise disjoint, define

$$F(x) = \begin{cases} F_m(x), & x \in B(a_m, R) \text{ for some } m, \\ 0, & x \notin \bigcup_{m=1}^{N} B(a_m, R). \end{cases}$$

Finally set $g(x) = T(F(x))$. Then $F(x) = x - a_m$ on $B(a_m, r)$, $F(x) = 0$ outside $\bigcup_m B(a_m, R)$, and $\|F(x)\| \leq R$. Hence g has support in U, $\sup_x \|g(x)\| \leq \|T\|_{\mathrm{op}} R \leq M$, and

$$Dg = T \quad \text{on} \quad H := \bigcup_{m=1}^{N} B(a_m, r).$$

Also $Dg = 0$ on $X \setminus U$, because g vanishes on a neighbourhood of that set.

It remains to estimate the Lipschitz constant of F. Fix m. Put $s = \|x - a_m\|$ and $t = \|y - a_m\|$. If $s \geq t$, since β is nonincreasing,

$$\begin{aligned} \|F_m(x) - F_m(y)\| &\leq \beta(s)\|x - y\| + |\beta(s) - \beta(t)|\, \|y - a_m\| \\ &\leq \|x - y\| + \frac{1}{L} t \log(s/t) \\ &\leq \left(1 + \frac{1}{L}\right) \|x - y\|, \end{aligned}$$

where the middle term is interpreted as 0 if $t = 0$, and the last inequality uses $t \log(s/t) \leq s - t \leq \|x - y\|$. The case $t \geq s$ is the same after writing

$$F_m(x) - F_m(y) = \beta(t)(x - y) + (\beta(s) - \beta(t))(x - a_m)$$

and using $s \log(t/s) \leq t - s \leq \|x - y\|$. Thus $\mathrm{Lip}(F_m) \leq 1 + 1/L$.

If x, y lie in the same ball $B(a_m, R)$, this gives the desired estimate for F. If one point lies in $B(a_m, R)$ and the other point lies outside all balls, the same estimate follows because F_m is zero outside $B(a_m, R)$. Finally, if $x \in B(a_m, R)$, $y \in B(a_n, R)$, and $m \neq n$, then

$$\|F(x) - F(y)\| \leq 2R \leq \|a_m - a_n\| - 2R \leq \|x - y\|.$$

Therefore $\mathrm{Lip}(F) \leq 1 + 1/L$, and so

$$\mathrm{Lip}(g) \leq \|T\|_{\mathrm{op}} \left(1 + \frac{1}{L}\right) \leq \|T\|_{\mathrm{op}} + \eta.$$

□

Remark 1. *The singleton theorem recorded in the previous version of this packet is the case $N = 1$ of the finite-set theorem.*

Corollary 1 (Pointwise non-optimality of $\mathfrak{C}(T)$)**.** *In dimensions at least 3, the bound $\mathfrak{C}(T)$ is not pointwise optimal for every individual instance of Lemma 4.4.*

Proof. By Remark 4.3(iii) of the source paper, there is a finite-dimensional normed space X of dimension at least 3 for which $\mathfrak{C}(\mathrm{Id}_X) > 1 = \|\mathrm{Id}_X\|$. Apply the finite-set theorem, in the singleton case, to $T = \mathrm{Id}_X$, $E = \{a\}$, and $\eta < \mathfrak{C}(\mathrm{Id}_X) - 1$. The resulting admissible map has Lipschitz constant strictly smaller than $\mathfrak{C}(\mathrm{Id}_X)$. Thus $\mathfrak{C}(T)$ cannot be the optimal constant for that individual instance. □

Corollary 2 (Optimal prescribed-derivative constant in rank at most two)**.** *Let X be finite-dimensional, let Y be a normed space, let $T \in \mathcal{L}(X, Y)$ have rank at most 2, and let $E \subseteq U \subseteq X$ be as in Lemma 4.4 of Dymond–Maleva, with E compact purely unrectifiable and U open. For every $\theta > 0$, the map $g : X \to T(X)$ in Lemma 4.4 can be chosen with*

$$\mathrm{Lip}(g) \leq \|T\|_{\mathrm{op}} + \theta.$$

If $E \neq \emptyset$, this is asymptotically sharp: any admissible g and open set H satisfying the lemma's derivative condition

$$\|Dg(x) - T\|_{\mathrm{op}} \leq \theta \quad \text{for Lebesgue almost every } x \in H$$

must obey

$$\mathrm{Lip}(g) \geq \|T\|_{\mathrm{op}} - \theta.$$

Proof. The upper bound is Lemma 4.4 of the source paper combined with the theorem above.

For the lower bound, suppose $E \neq \emptyset$. Then the open set H appearing in Lemma 4.4 is nonempty, so it has positive Lebesgue measure in the finite-dimensional space X. At almost every $x \in H$, the derivative condition gives

$$\|Dg(x)\|_{\mathrm{op}} \geq \|T\|_{\mathrm{op}} - \|Dg(x) - T\|_{\mathrm{op}} \geq \|T\|_{\mathrm{op}} - \theta.$$

For every point where the derivative exists, $\|Dg(x)\|_{\mathrm{op}} \leq \mathrm{Lip}(g)$. Thus $\mathrm{Lip}(g) \geq \|T\|_{\mathrm{op}} - \theta$. □

Limitations and Novelty Check

This is a partial result. It does not settle the general rank-three-and-higher case for arbitrary compact purely unrectifiable sets. The finite-set theorem shows, however, that $\mathfrak{C}(T)$ is not a pointwise optimal constant for every individual instance. Any genuine obstruction to the hoped-for constant $\|T\|_{\mathrm{op}}$ must therefore use infinite, nontrivial purely unrectifiable sets, not finite sets.

Local duplicate checks over the run registry, solution index, attempt index, and proof-gap index found no prior packet for arXiv:2504.04117, the cylinder constant $\mathfrak{C}(T)$, the rank-two subcase, finite-set prescribed derivatives, or the singleton pointwise non-optimality observation. Web searches on 2026-06-20 for exact phrases including “cyl(T)”, “It would be of interest to determine whether this upper bound is optimal”, “monotone basis three-dimensional Banach space”, and the paper title found the arXiv source page and background on monotone bases but no later answer to this optimality question.

References

Michael Dymond and Olga Maleva, *Extreme non-differentiability of typical Lipschitz mappings*, arXiv:2504.04117, 2025.